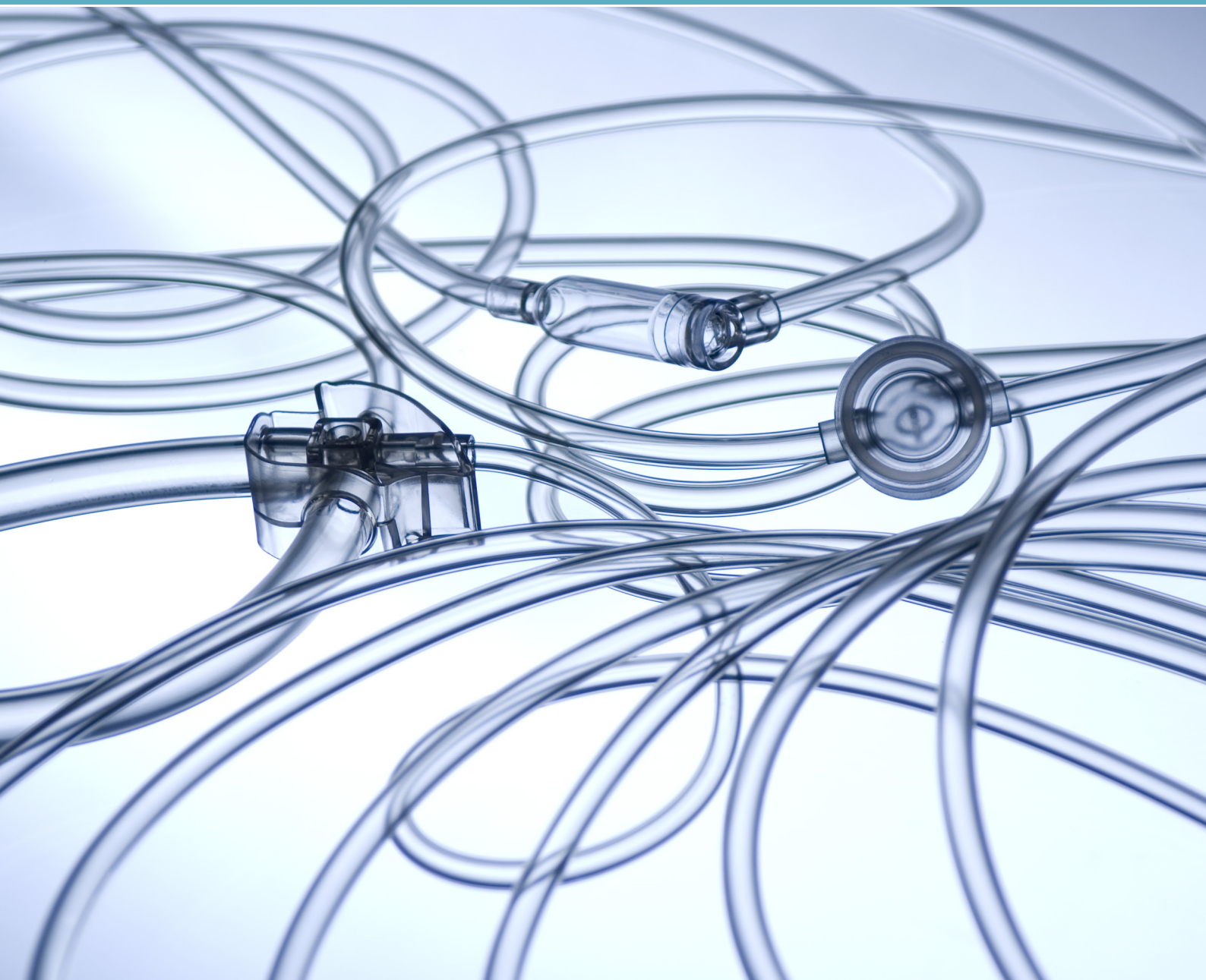


Haemodialysis

Bloodlines, Tubing Systems, Accessories

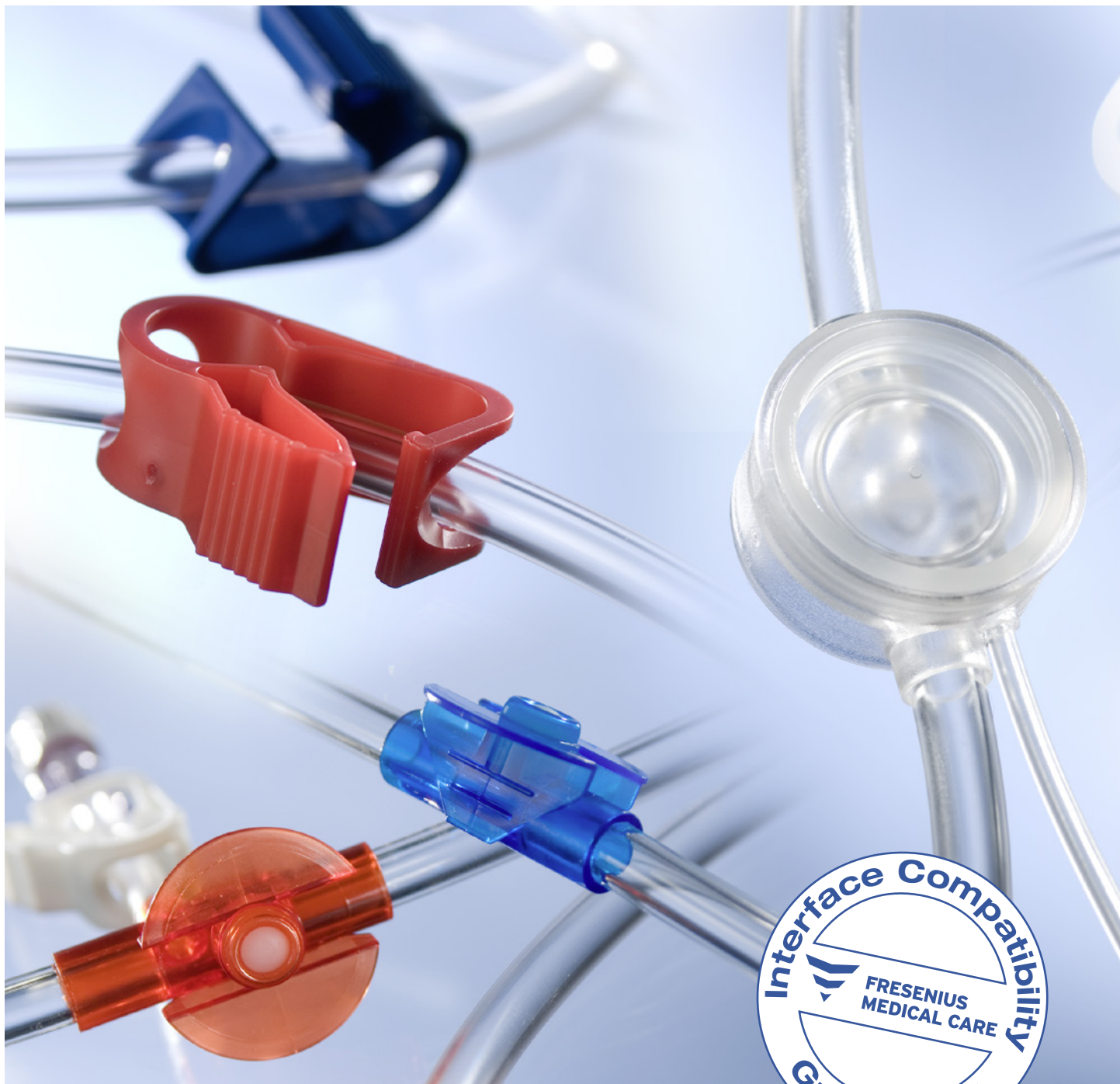
Product Range



**FRESENIUS
MEDICAL CARE**

Ensuring safety with the authentic Fresenius Medical Care Original Bloodlines ...

Quality and reliability characterise the Fresenius bloodlines in function and use.



Handling

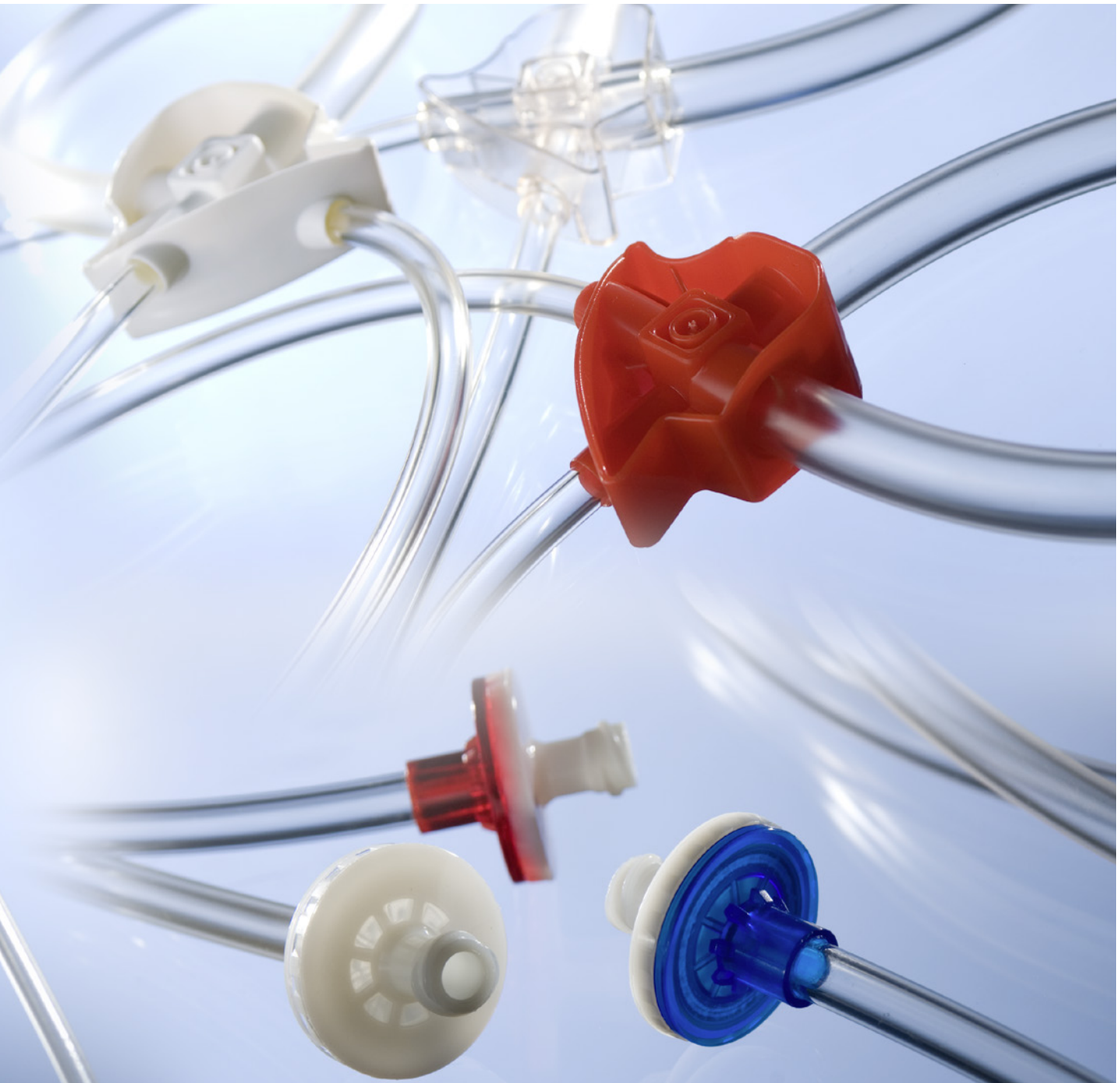
- Clearly colour-coded components
- Ergonomic components and layout

*We guarantee that all our bloodlines are compatible with all software and hardware versions of our respective therapy systems.

Safety

- Excellent safety and performance with our 4008 and 5008 machines
- Transducer protector with integrated inspection windows
- Injection port with enlarged finger-protection shield

... developed to meet your needs



Quality

- Certified quality systems for development, production and sales
- Perfect compatibility to machine interfaces

Biocompatibility

- Use of plasticisers with high haemocompatibility
- Free of DEHP

5008 Therapy System – Beyond your expectations

Enabling the application of HighVolumeHDF® in the most practical way.



ONLINE Priming/HighVolumeHDF®

- Saline free priming of extracorporeal circuit
- HighVolumeHDF® treatment as standard
- Enabled by the SafeLine included in every bloodline set

Reliable Precision

- Exact measurement of arterial pressure via pressure dome
- Trouble-free air detection directly on system tube

Convenience

- Clear layout and optimised tube lengths
- Machine assisted set-up and disassembling
- Reduced filling volume and less blood-air contact

Product Range 5008

DEHP-free

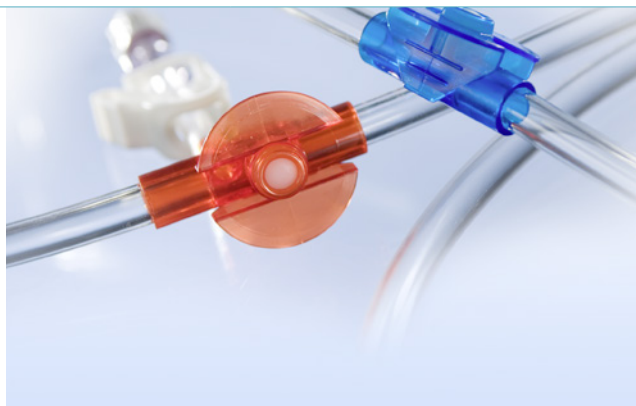
LifeLine_{Beta}

- Phtalate free
- Beta-sterilisation by accelerated electrons (e-beam)

Adult Bloodlines

Beta sterilisation – DEHP free

Type	To be used with	Filling volume	Units per box	Art.-No.
AV-Sets				
AV-Set ONLINEplus 5008-R	5008/5008S	132 mL	20	F00000384
AV-Set ONLINEplus BVM 5008-R	5008	136 mL	20	F00000385
AV-Set ONLINE Priming 5008S-R	5008S	132 mL	20	F00000700
AV-Set 5008 CorDiax post-R	5008/5008S	132 mL	20	F00003267
SN-Sets				
SN-Set ONLINEplus 5008-R	5008/5008S	193 mL	14	F00000386
SN-Set ONLINEplus BVM 5008-R	5008	196 mL	14	F00000387



Paediatric Bloodlines

Beta sterilisation – DEHP free

Type	To be used with	Filling volume	Units per box	Art.-No.
AV-Sets				
AV-Set ONLINEplus BVM Paed 5008-R	5008	114 mL	20	F00001068
SN-Sets				
SN-Set ONLINEplus BVM Paed 5008-R	5008	169 mL	14	F00001069



BVM: Blood Volume Monitor
-R: Beta Sterilisation (Radiation)

Inner diameter of pump segment: 8 mm

Diameter of venous chamber: 22 mm

4008 – Sustained dependability

Reliability established over several years and millions of treatments.



Superior quality and safety

- Consistent and validated manufacturing
- Rigorous testing of bloodline performance
- Highly compatible interface between bloodlines and machine

Reliable therapy

- Matured technology recognised worldwide
- Extensive experience in the production of bloodlines – ensuring consistent product quality

Meeting diverse needs

- Appropriate sterilisation to suit varying demands
- Broad product range to meet all requirements

Product Range 4008

DEHP-free

LifeLine_{Beta}

- Phtalate free
- Sterilisation by accelerated electrons (e-beam)



Adult Bloodlines

Beta sterilisation – DEHP free

Type	To be used with	Inner diameter pump segment	Filling volume	Units per box	Art.-No.
AV-Sets					
AV-Set B-R	4008	8 mm	161 mL	20	F00001124
AV-Set SRB-R	4008	8 mm	161 mL	20	F00000257
AV-Set SRB BVM-R	4008	8 mm	172 mL	20	F00000258
AV-Set FMC (FA 204 C/FV 204 C)-R	4008	8 mm	161 mL	20	F00004757
SN-Sets					
SN-Set SRB L-R	4008	8 mm	211 mL	14	F00000259

Paediatric Bloodlines

Beta sterilisation – DEHP free



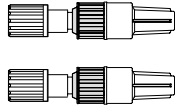
Type	To be used with	Inner diameter pump segment	Filling volume	Units per box	Art.-No.
AV-Sets					
AV-Set-FMC Paed R	4008	6.4 mm	117 mL	20	F00001064
AV-Set-FMC BVM Paed R	4008	6.4 mm	126 mL	20	F00001065
AV-Set-FMC Paed/Baby R	4008	6.4 mm	56 mL	26	F00001063
SN-Sets					
AV-SN-Set-FMC Paed R	4008	6.4 mm	155 mL	20	F00001067
AV-SN-Set-FMC Paed/Baby R	4008	6.4 mm	88 mL	20	F00001066

All paediatric bloodlines 4008 are equipped with spike, waste bag and recirculation connector.

- S: Spike
 R: Recirculation connector
 B: Waste bag
 L: Long tubes to SN expansion chamber
 (needed for SN-treatment with 4008 S)
 BVM: Blood Volume Monitor
 -R: Beta Sterilisation (Radiation)

Diameter of venous chamber: 22 mm

Components



Patient connectors

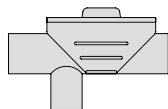
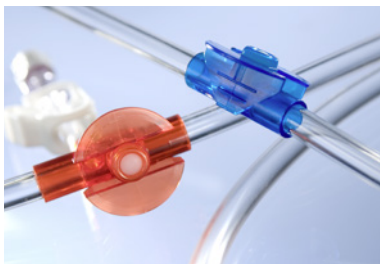
Optimal handling and safety:

- Ergonomic grip on the rotating part ensures secure connection
- Easy to remove protective caps



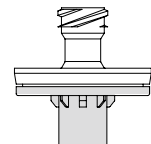
Line clamps

- Colour coding
- Optimal ergonomics for easy and reliable handling
- No additional clamps required



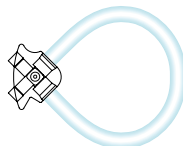
Injection ports

- Latex-free septum
- Colour coding
- Large finger-protection shield



Transducer protectors

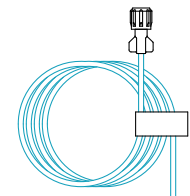
- Inspection windows
- Colour coding



Pump segment

Perfect fit to the Fresenius Medical Care dialysis machines (geometry, shore hardness, occlusion performance)

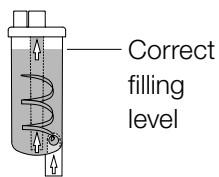
- Qualified for the effective blood flow measurement
- Allows achievement of prescribed dialysis dose



Heparin line

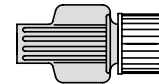
- Luer-Lock female Adapter with protecting cap
- Easy to remove paper tape

Components



Arterial bubble catcher

- Dynamic blood flow due to special geometry
- Safe and effective air separation



Dialyser connector

- Soft material allows safe connection to dialyser
- Easy to remove protective caps



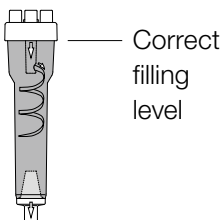
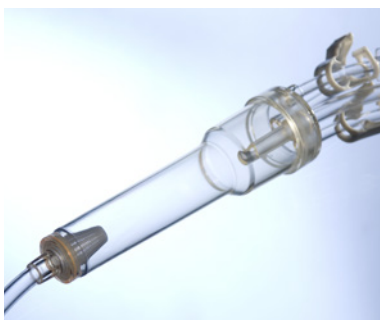
Single Needle chamber

- Optimised design
- Dynamic, controlled bloodflow
- Easy handling and protection of the pressure line



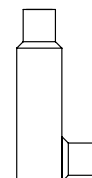
Pressure dome

- Precise pressure measurement
- Reduced filling volume
- No blood-air contact



Venous drip chamber

- Inlet distributor reduces activation of clotting
 - Directed blood flow
 - No "free fall" of blood
- Allows an easy and correct positioning in the air detector due to the special shape of venous bubble trap (22 mm Ø)



BVM cuvette

- Precise BVM measurement
- Easy handling

AV-Set ONLINEplus 5008-R (F00000384)

Arterial line

Venous line

SafeLine

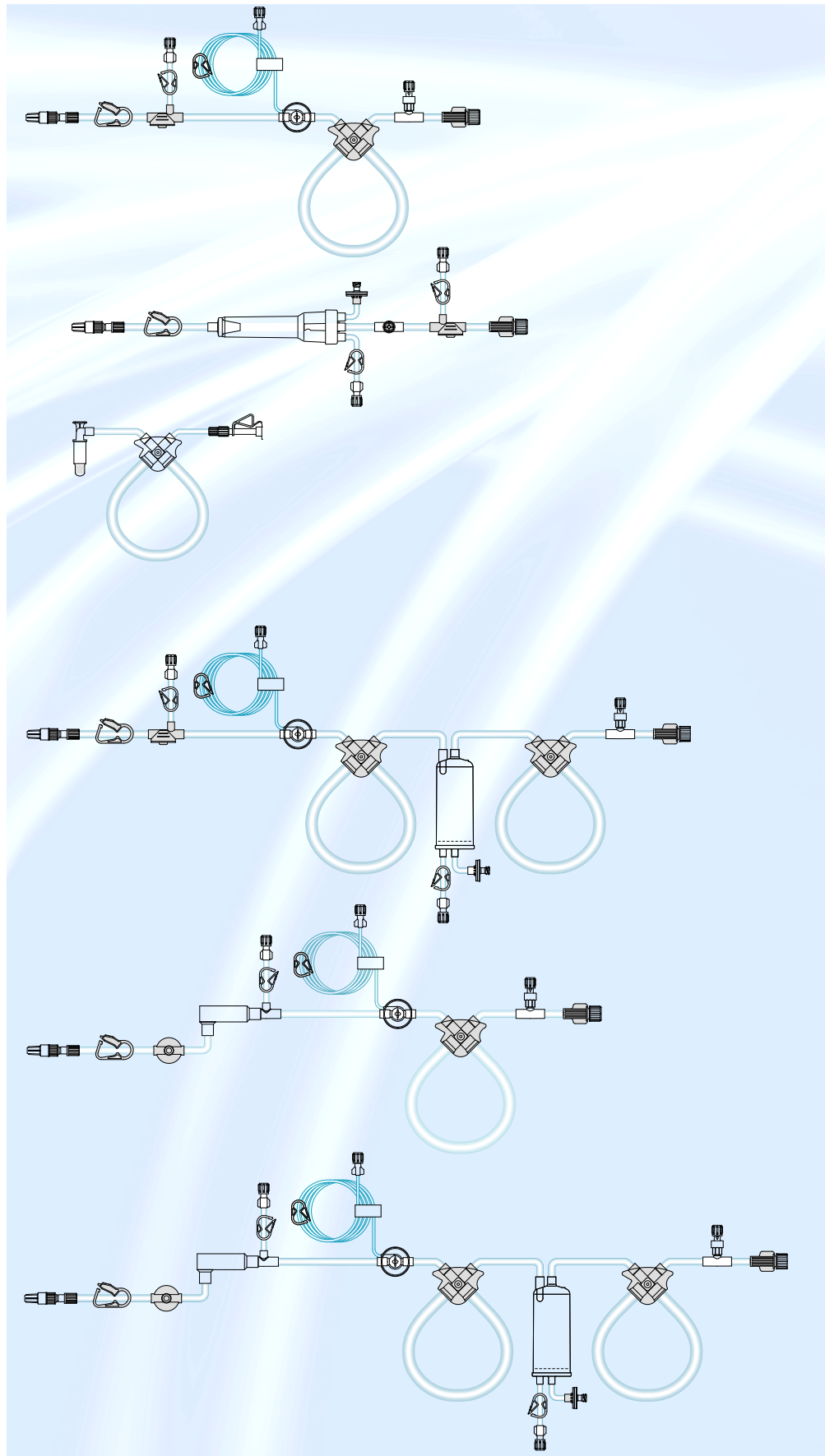
Variants

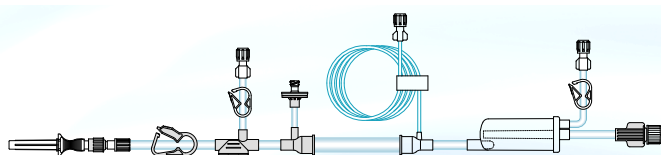
(Venous line and SafeLine
unchanged)

SN-Set ONLINEplus 5008-R
(F00000386)

AV-Set ONLINEplus BVM 5008-R
(F00000385)

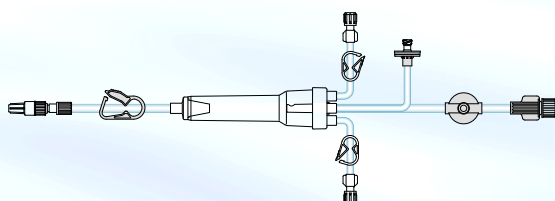
SN-Set ONLINEplus BVM 5008-R
(F00000387)



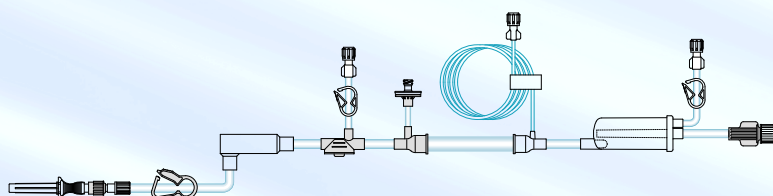


AV-Set SRB-R (F00000257)

Arterial line



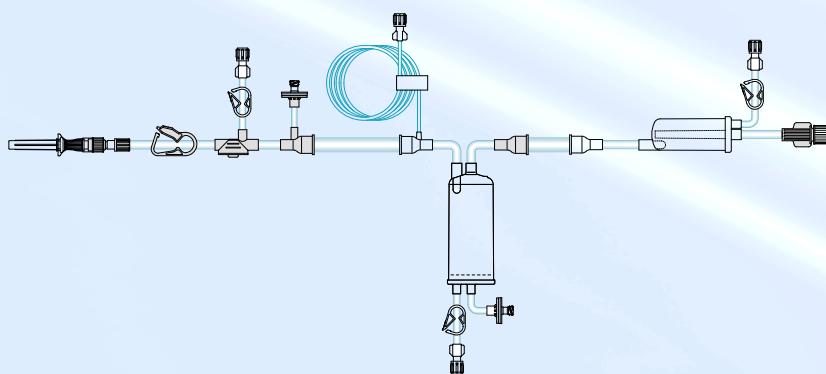
Venous line*



Variants

(Venous lines unchanged)*

AV-Set SRB BVM-R
(F00000258)



SN-Set SRB LR
(F00000259)

* Delivered with a drainage bag

Technical drawings paediatric lines

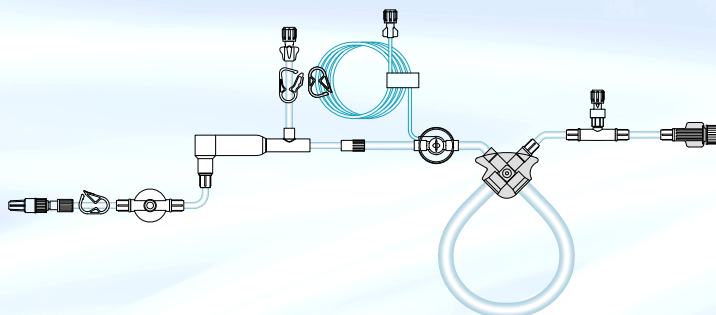
5008

AV-Set ONLINEplus – BVM

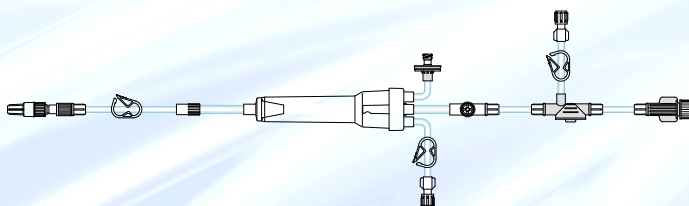
Paed 5008-R

(F00001068)

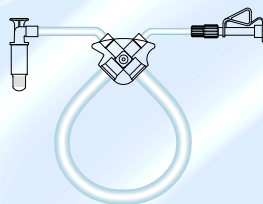
Arterial line



Venous line



SafeLine

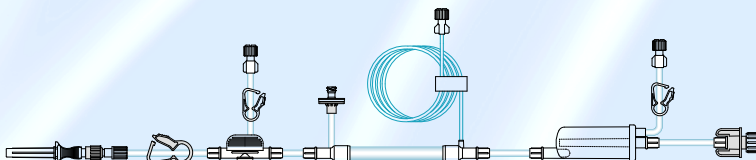


4008

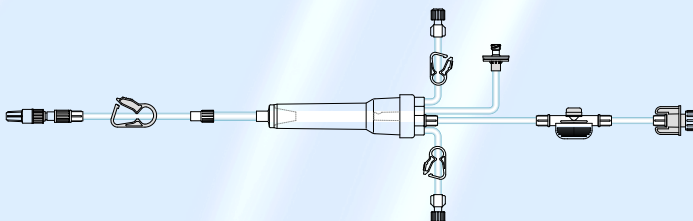
AV-Set-FMC Paed R

(F00001064)

Arterial line

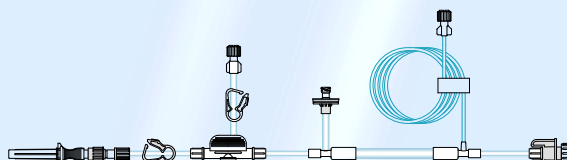


Venous line*

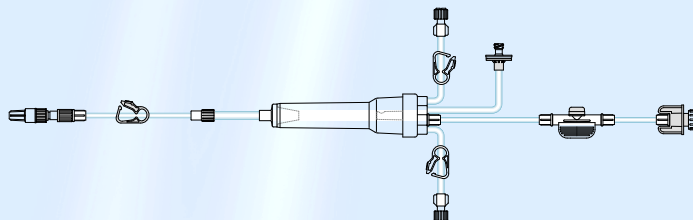


AV-Set-FMC Paed/Baby R
(F00001063)

Arterial line



Venous line*



*Delivered with a drainage bag

Important handling aspects

Pressure line

Connect the pressure transducer protector to the machine ensuring it is straight and secure (do not overtighten!). If the membrane is wet, replace either the bloodline or the pressure line with a new one.

Arterial bubble catcher

For level setting release the cap, raise the level slowly to the filling line. Make sure that the caps are secure and the clamps closed.

Venous bubble catcher (4008)

Rest the shoulder of the chamber on the top of the level detector holder, so the filter of the chamber is 1 cm below the level of the door.

Make sure that the caps are secure and the clamps closed.

Pump segment (4008)

Insert the pump segment with the red colour coding on the left hand side.

SN-pump segment (4008)

Make sure that all air has been removed during the priming of the pump segment.

Injection port

Clean the septum by using a disinfectant. Hold the injection port by the finger grips underneath the protection shield. Insert the needle directly into the septum (90°).

SN-chamber

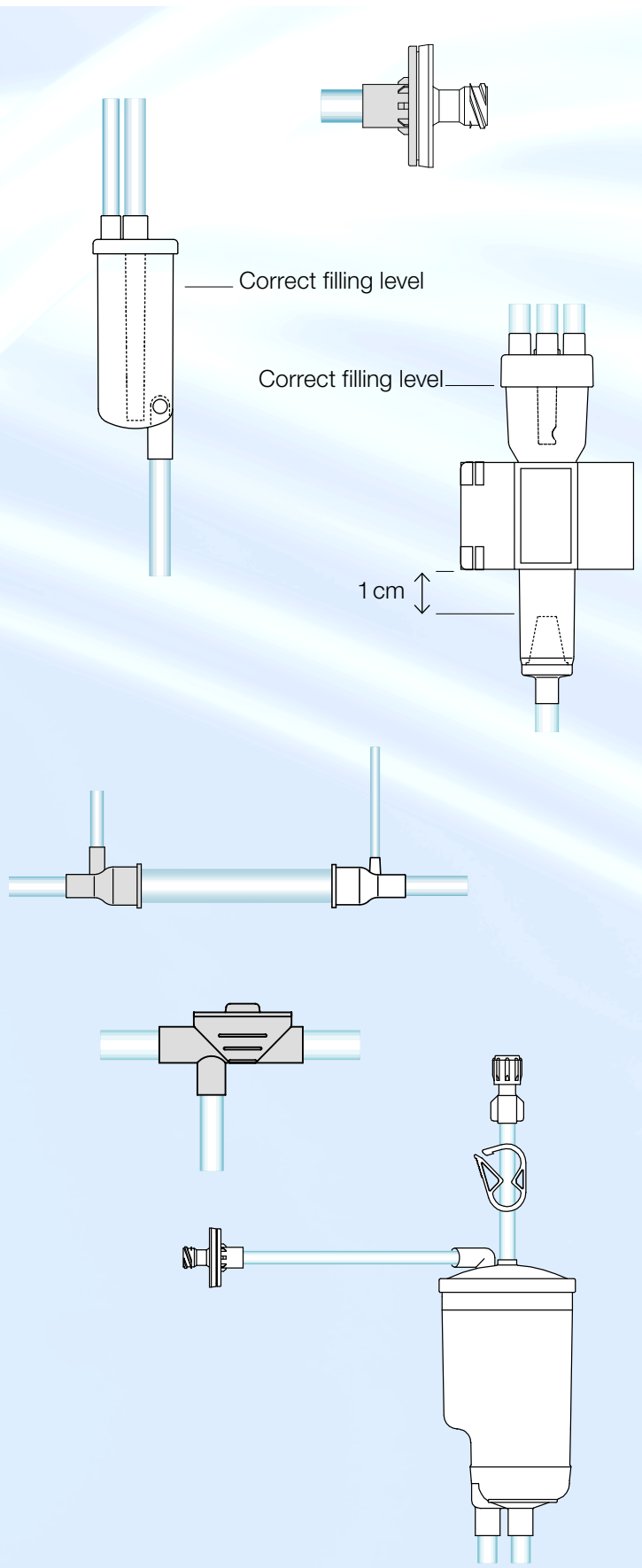
Place the chamber into the holder.

Connect the pressure line (see above).

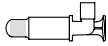
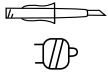
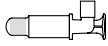
Make sure that the caps are secure and the clamps closed.

Do not invert the chamber.

During priming, the filling level will be set automatically (do not raise the level manually).



Accessories

Type	To be used with	Sterilisation	Description	Connectors	Units per box	Art.-No.
HF/HDF						
Drainage connector	4008/5008	Beta	Connector used for hemodialysis circuits		12	7030031
Safeline	4008	Steam	Set to be used for performing on-line treatment with 4008 bloodlines		100	5046801
Infusion sets						
Frekaflex infusion set	4008	ETO	Infusion system with roller clamps and drip chamber		150	2889011
Dialysis accessories	4008	ETO	Infusion set with spike		130	5017151
Waste bags						
Waste bag 2,000 mL	4008	ETO	Collection bag, 2000 ml with female Luer-Lock connector		100	5015091
Transducer protectors						
Transducer protector	4008	ETO	Pressure monitor. The device is intended to be assembled on the transducer protector present on the 4008 bloodlines (arterial/venous and single needle pressure monitoring) as additional protection		100	F00007580
Pressure line 30 cm	4008/5008	ETO	Complete pressure measurement line with filter, male Luer-Lock connector, 30 cm, blue		100	5014631
Various accessories						
Heparin syringe 30 mL	5008	ETO	Luer-Lock male, separate closure cap, latex-free plug		100	5030321
Heparin syringe 20 mL	4008/5008	ETO	Luer-Lock male, latex-free plug		120	F00001255
SN-adaptor Luer-Lock	4008/5008	ETO	Y-Adapter for single needle treatment		100	5027851
Scissor clamp	4008/5008	---	For clamping tubes		5	2845241
Recirculation connector	4008/5008	Beta	Tube adapter with 2 female Luer-Lock connectors and grommet It is intended to be used in combination with Fresenius bloodlines and tubing system for filling / emptying		100	F00007581
Spike	4008/5008	Beta	Spike connector Luer-Lock female. It is intended to be connected to the luer lock male (patient connector) for the priming with bag. To be used in combination with Fresenius bloodlines		100	F00007582
Dialyser exchange set	4008/5008	ETO	Tubing system for exchange of dialyser during treatment, short-circuit AV-set (e.g.: change Haemodialyzer)		75	5018001
Fluid sampling set	5008	Beta	5008 Accessory used to withdraw fluid from the machine 5008 in order to test the quality of the solution prepared by the machine.		15	F00006222

Type	Sterilisation	Description	Units per box	Art.-No.
HD Connection-Disconnection Set				
proHD Set S	ETO	All components for connection and disconnection of fistula patients (AVF/AVG), including drape, nitrile gloves, non-woven gauzes, adhesive tapes, haemostatic plasters	120	F00001621
proHD Set M	ETO	All components for connection and disconnection of fistula patients (AVF/AVG), including drape, nitrile gloves, non-woven gauzes, adhesive tapes, haemostatic plasters	120	F00001622
proHD Set L	ETO	All components for connection and disconnection of fistula patients (AVF/AVG), including drape, nitrile gloves, non-woven gauzes, adhesive tapes, haemostatic plasters	120	F00001623
CVC Connection Set*				
proHD CVC Set S – Connection Set	ETO	All components for connection of CVC patients, including gloves, drapes, swabs, syringes, tape strips	40	TMD000081
proHD CVC Set M – Connection Set	ETO	All components for connection of CVC patients, including gloves, drapes, swabs, syringes, tape strips	40	TMD000091
proHD CVC Set L – Connection Set	ETO	All components for connection of CVC patients, including gloves, drapes, swabs, syringes, tape strips	40	TMD000101
CVC Disconnection Set*				
proHD CVC Set S – Disconnection Set	ETO	All components for disconnection of CVC patients, including gloves, drapes, swabs, syringes, caps, tape strips	44	TMD000111
proHD CVC Set M – Disconnection Set	ETO	All components for disconnection of CVC patients, including gloves, drapes, swabs, syringes, caps, tape strips	44	TMD000121
proHD CVC Set L – Disconnection Set	ETO	All components for disconnection of CVC patients, including gloves, drapes, swabs, syringes, caps, tape strips	44	TMD000131

*For one treatment 1 connection and 1 disconnection set needed.

The letters S, M and L in the name of the proHD Sets and proHD CVC Set indicate the glove size.



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MEDICAL CARE**

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